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/*  
  
    Name: Andrew Castro  
    Date: 9/7/2023  
    Purpose: Calculate the cost of filling a pool with given dimensions of pool  
  
*/  
  
#include <iostream>  
  
using namespace std;  
  
int main()  
{  
    //Message to user to enter a pool's length, width and depth in feet measurement  
    cout << "Enter the pool's length, width and depth in feet:" << endl;  
  
    //Holds input for length in feet  
    double length;  
  
    //Holds input for width in feet  
    double width;  
  
    //Holds input for depth in feet  
    double depth;  
  
    //Holds value of pool's volume in cubic feet  
    double poolVolume;  
  
    //Holds value of  
    double waterVolume;  
  
    //Takes input from user for length, width, and depth  
    cin >> length >> width >> depth;  
  
    //Calculate pool's volume  
    poolVolume = length * width * depth;  
  
    //Prints out given length, width, and depth  
    cout << "Length\t Width\t Depth" << endl;  
    cout << length << "\t " << width << " \t " << depth << endl;  
  
    //Prints out pool volume  
    cout << "Pool Volume is:\n" << poolVolume << " cubic feet" << endl;  
  
    //Checks if depth is greater than 3 inches (0.25 of a foot is 3 inches) for  
        filling pool requirement
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if (depth > 0.25 )
{
    //Subtracts three inches from given depth for filling pool
    depth = ((depth * 12) - 3)/12;

    //Calculates the water volume to fill pool
    double waterVolume = length * width * depth;

    //Prints out water volume of pool
    cout << "Water Volume of Pool:\n" << waterVolume << " cubic feet" << 
    endl;

    //Prints out the cost to fill pool
    cout << "Cost to Fill Pool: \n$" << (waterVolume * 0.77 + 100.00) << 
    endl;

}
else
{
    waterVolume = 0;
    //Tells user water volume is 0
    cout << "Water Volume of Pool: \n" << waterVolume << endl;

    //Tells user the pool cannot be filled
    cout << "Pool cannot be filled " << endl;

    //Tells user no cost to fill pool since it cannot be filled
    cout << "Cost to Fill Pool: \n$" << 0 << endl;
}

//Prints out Author's name
cout << "Andrew Castro" << endl;
}
```